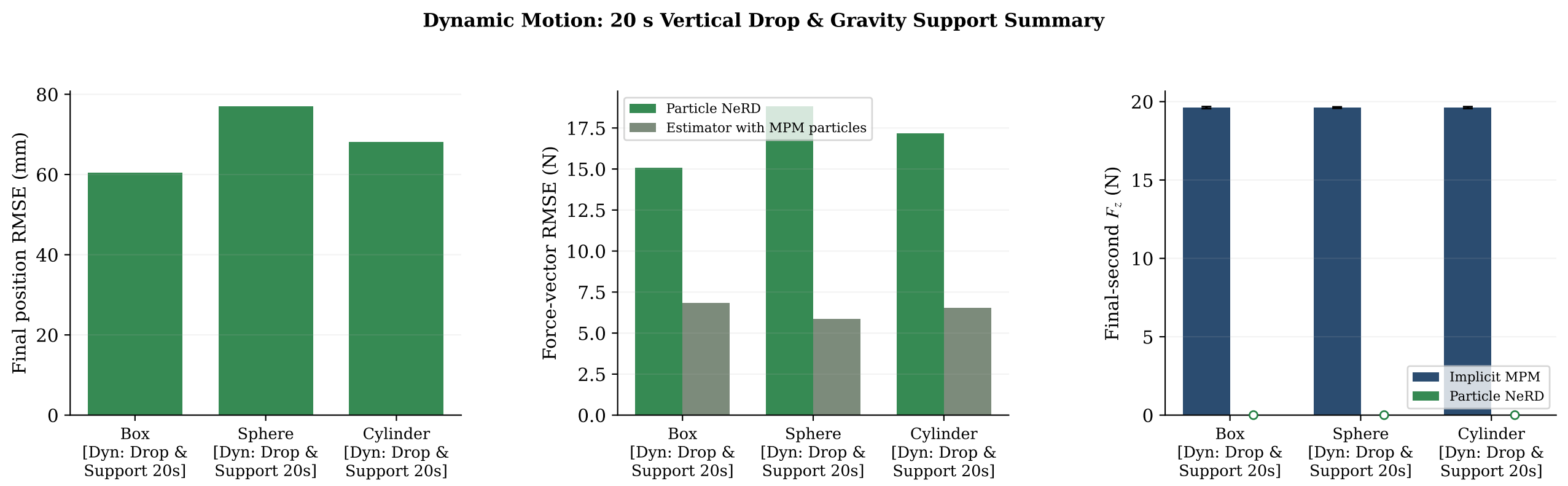

Figure 6: Frozen object-model checkpoint 11,000. Dynamic Motion: 20 s Drop & Gravity Support for Box, Sphere, and Cylinder. Left: terminal particle-position RMSE. Center: force-vector RMSE. Right: mean Fz during the final second (19–20 s).